

EV SALES ANALYSIS SQL ANALYSIS WITH SCREENSHOTS

Welcome

SQLQuery_1 - (58) l...db (SA)

Run Cancel Disconnect Change Database: EV_db Estimated Plan Enable Actual Plan Parse Enable SQLCMD To Notebook

```
10 -- Calculate the total number of electric vehicles (both BEV and PHEV)
11 WITH ElectricVehicles AS (
12     SELECT COUNT(*) AS electric_count
13     FROM WA_EV
14     WHERE electric_vehicle_type IN ('Battery Electric Vehicle (BEV)')
15 ),
16
17 -- Calculate the total number of vehicles
18 TotalVehicles AS (
19     SELECT COUNT(*) AS total_count
20     FROM WA_EV
21 )
22
23 -- Calculate the EV adoption rate
24 SELECT
25     (CAST(ElectricVehicles.electric_count AS FLOAT) / TotalVehicles.total_count) * 100 AS complete_ev_adoption_rate
26 FROM
27     ElectricVehicles, TotalVehicles;
28
```

Results Messages

	complete_ev_adoption_rate
1	78.45916979924819

PROBLEMS OUTPUT TERMINAL TASKS

No task history to display.

Screen Reader Optimized Ln 27, Col 37 Spaces: 4 UTF-8 LF 1 rows MSSQL 00:00:00 localhost: EV_db (58)

```

4
5  --Number of Electric Vehicles only by County
6  SELECT county, COUNT(*) AS electric_vehicle_count
7  FROM WA_EV
8  WHERE electric_vehicle_type='Battery Electric Vehicle (BEV)'
9  GROUP BY county
10 ORDER BY electric_vehicle_count DESC;
11
12

```

Results Messages

	county	electric_vehicle_count
1	King	83114
2	Snohomish	19828
3	Pierce	12090
4	Clark	8791
5	Thurston	5484
6	Kitsap	4975
7	Spokane	3608
8	Whatcom	3590
9	Benton	1742
10	Skagit	1621
11	Island	1587
12	Chelan	953
13	Yakima	842
14	Clallam	819
15	San Juan	797
16	Jefferson	787
17	Cowlitz	747
18	Mason	670
19	Kittitas	581
20	Lewis	498
21	Franklin	493
22	Grays Ha...	489
23	Grant	483
24	Walla Wa...	336
25	Douglas	313
26	Whitman	255
27	Klickitat	241
28	Okanogan	208
29	Stevens	159
30	Skamania	157
31	Pacific	127

```

10
17 --Average Electric Range by Vehicle Make,model
18 SELECT make,model, AVG(electric_range) AS average_electric_range
19 FROM WA_EV
20 GROUP BY make,model
21 ORDER BY average_electric_range DESC;
22
23

```

Results Messages

	make	model	average_electric_range
1	HYUNDAI	KONA	258
2	TESLA	ROADSTER	228
3	JAGUAR	I-PACE	204
4	TESLA	MODEL S	173
5	CHEVROLET	BOLT EV	170
6	AUDI	E-TRON	129
7	TESLA	MODEL X	129
8	VOLKSWAGEN	E-GOLF	107
9	TESLA	MODEL 3	104
10	TOYOTA	RAV4	102
11	THINK	CITY	100
12	WHEEGO ELECTRIC CARS	WHEEGO	100
13	KIA	SOUL EV	97
14	KIA	SOUL	93
15	AUDI	E-TRON SPORTBACK	91
16	BMW	I3	90
17	MERCEDES-BENZ	B-CLASS	87
18	FIAT	500	85
19	CHEVROLET	SPARK	82
20	NISSAN	LEAF	82
21	FORD	FOCUS	79
22	HYUNDAI	IONIQ	71
23	KIA	NIRO	68
24	PORSCHE	TAYCAN	65
25	FORD	RANGER	64
26	SMART	FORTWO ELECTRIC DRIVE	63
27	SMART	FORTWO	62
28	MITSUBISHI	I-MIEV	61
29	SMART	EQ FORTWO	58
30	AZURE DYNAMICS	TRANSIT CONNECT ELECTRIC	56
31	HONDA	CLARITY	47

```

45 --Market Share of Each Vehicle Make in the EV Market
46 WITH BatteryEVs AS (
47     SELECT COUNT(*) AS total_battery_ev_count
48     FROM WA_EV
49     WHERE electric_vehicle_type= 'Battery Electric Vehicle (BEV)'
50 ),
51
52 MakeCounts AS (
53     SELECT make, COUNT(*) AS make_count
54     FROM WA_EV
55     WHERE electric_vehicle_type='Battery Electric Vehicle (BEV)'
56     group by make
57 )
58 )
59
60 SELECT m.make, round((m.make_count * 100.0 / t.total_battery_ev_count),2,2) AS market_share_percentage
61 FROM MakeCounts m
62 JOIN BatteryEVs t on 1=1
63 ORDER BY market_share_percentage DESC;
64
65

```

Results Messages

	make	market_share_percentage
1	TESLA	56.090000000000
2	NISSAN	9.190000000000
3	CHEVROLET	6.350000000000
4	FORD	4.330000000000
5	KIA	3.880000000000
6	VOLKSWAGEN	3.540000000000
7	RIVIAN	3.310000000000
8	HYUNDAI	2.950000000000
9	BMW	1.940000000000
10	AUDI	1.600000000000
11	VOLVO	0.970000000000
12	MERCEDES-BENZ	0.930000000000
13	POLESTAR	0.700000000000
14	SUBARU	0.700000000000
15	TOYOTA	0.500000000000
16	MINI	0.500000000000
17	FIAT	0.500000000000
18	PORSCHE	0.440000000000
19	CADILLAC	0.380000000000
20	LEXUS	0.200000000000
21	LUCID	0.170000000000
22	SMART	0.160000000000

```

69  --Analysis of Vehicle Age vs. Electric Range
70  SELECT vehicle_age, AVG(electric_range) AS avg_electric_range
71  FROM WA_EV
72  GROUP BY vehicle_age
73  ORDER BY avg_electric_range DESC;
74
75
76
77
78
79  |
80
81
82
83
84

```

Results Messages

	vehicle_age	avg_electric_range
1	4	237
2	14	226
3	16	210
4	5	176
5	6	156
6	7	114
7	8	102
8	9	97
9	21	95
10	22	95
11	10	80
12	11	79
13	25	74
14	13	70
15	12	60
16	24	58
17	27	39